

Tzu-Han Lin

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PROFESSIONAL SUMMARY

A detail-oriented Software Developer with a Master's in Computer Information Systems from Kean University. Experienced in full-stack development, data analysis, and machine learning, I am driven to solve complex problems by developing innovative software solutions.

AWARDS

Dean List | Kean University

Dec 2022

Summer Research Grant: Students Partnering with Faculty (SpF) | Kean University

May 2022, 2023

TECHNICAL SKILLS

Languages : C#, Python, php, Java, SQL (MS-SQL, MySQL), PL/SQL, JavaScript, HTML/CSS

Web Frameworks : ASP.NET (.NET Core), React, Flask, Django

DevOps and API Tools : Git, Docker, Jira

Cloud and Server configuration : Azure Cloud, SQL Server, Microsoft IIS, Linux

Deep Learning frameworks : Pytorch

EXPERIENCE

Graduate Assistant: Software Developer | Kean University

Sep 2023 – Present

- Developed a full-stack system with ASP.NET for IPEDS reporting, incorporating annual data storage and inter-year validation, significantly reducing inter-departmental communication costs
- Implemented a comprehensive front-end validation system using ReactJS, enhancing user experience with datatype checks, explanatory tooltips, and navigation buttons across scattered input fields.
- Designed RESTful APIs to optimize back-end processes, including server-side validation and database interactions.

Programming Tutor | Kean University

Sep 2022 – May 2023

- Experienced in guiding students through difficulties in MySQL, PHP, data structures, Java, and Python.

EDUCATION

Kean University

MS in Computer Information Systems | GPA : 4.0

Jan 2022 – present

National Tsing Hua University

BS in Medical Science

Sep 2013 – Jun 2017

RESEARCH

Selective Image-Based Weakly Supervised Segmentation in Breast Ultrasound

May 2023 – Jan 2024

- Implemented a weakly supervised image segmentation framework, integrating an innovative auto-selection algorithm for identifying high-potential candidates from class activation map. This approach achieves an Intersection over Union (IoU) score that is 82.9% of what is attained by fully supervised methods
- Accepted by the 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, 2024

PROJECTS

TideTrack: Harmful algal bloom observing and forecasting system | React, Python, Django, MySQL, Azure Jan-Apr 2024

- Designed and developed TideTrack, a system to perform ETL (Extract, Transform, Load) operations on diverse data sources and forecast the intensity of harmful algal blooms in the Gulf of Mexico, utilizing machine learning model and weather forecast data to provide timely environmental insights.

The Virtual Try-on Booth | Flask, Python, Docker

Sep-Dec 2023

- Designed API for an e-commerce clothing website, integrating deep learning models to enable an image-based virtual try-on feature.
- Authored Docker files and deployed various components on the server, ensuring scalable and efficient operation of the virtual try-on functionality.

Inventory Management and Order Processing System | HTML/CSS, PHP, MySQL, Linux

Sep-Dec 2023

- Designed an online inventory and order management system utilizing concurrent transactions to prevent conflicts and ensure data consistency across multiple user interactions, improving system reliability and user experience.
- Implemented database triggers to enhance auditing capabilities, ensuring robust monitoring and historical tracking of table modifications for improved operational accountability.

Mars Terrain Type Segmentation with Novel Edge Detection | Python, PyTorch

Jan-May 2023

- Performed comparative analysis between a Fully Convolutional Network (FCN) and a custom FCN with edge feature channel for image segmentation, focusing on Mars terrain type.
- Improved the bedrock type recall rate by 5%.

Analysis of job hiring trends across diverse categories during the pandemic | Python, Pandas

Jan-May 2023

- Conducted an analysis of job hiring trends during the pandemic across various categories by developing a web scraper to collect data from LinkedIn. Utilized Python and Pandas for data manipulation and created line charts to visualize and analyze the trends over time.

Online Banking System | HTML/CSS, PHP, MySQL, Linux

Sep-Dec 2022

- Developed an online banking system with a three-tier architecture, enabling secure user login, seamless transaction processing, and data validation process.